COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 3 – FALL 2025

Course Title:	Digital System Design			Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais			Program Name:	BS Computer Engineering		
Semester:	7^{th}	Batch:	FA22	Section:	A, B	Date Given:	5^{th} December, 2025
Submission Date:	9^{th} December, 2025			Maximum Marks:		20	
Name:				Registration Number:			

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- **No late submission**.

Question 1: [CLO2-PLO3-C5] [10 Marks]

Use the Implication Table method to optimize the Sequential Logic by reducing the number of states for the FSM shown in Figure 1 and design its circuit at gate level. Briefly summarize the optimization you obtained.

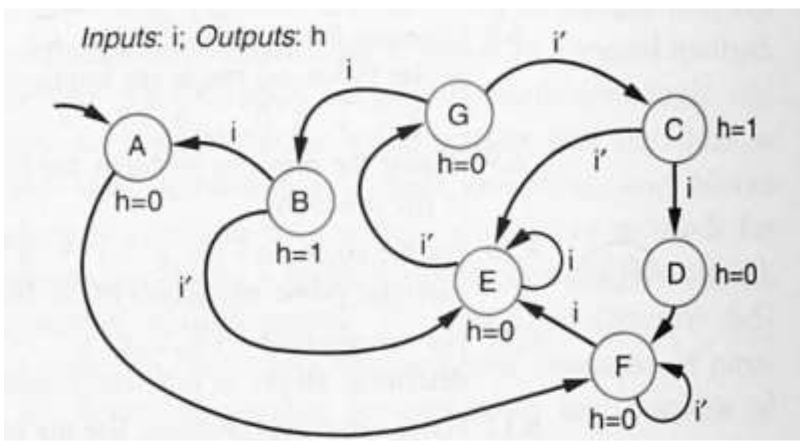
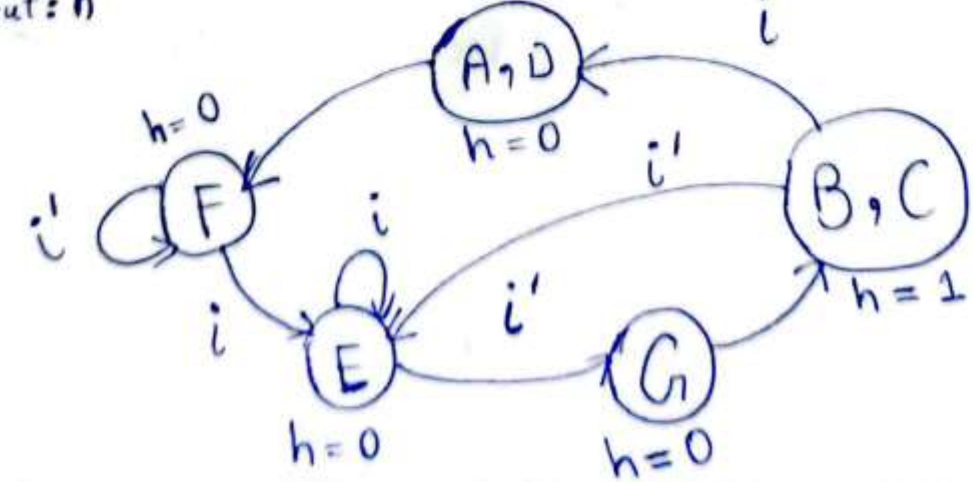


Figure 1

	A	B	C	D	E	F
B	X					
C	X	A,D E,E				
D	F,F F,F	X	X			
E	~~F,E F,G~~	X	X	~~F,E F,G~~		
F	~~F,E F,F~~	X	X	~~F,E F,F~~	~~G,F E,E~~	
G	~~F,C F,B~~	X	X	~~F,C F,B~~	~~E,B G,C~~	~~E,C E,B~~

input: i
output: h



Question 2: [CLO2-PLO3-C5] [10 Marks]

Design a 16-bit multiplier using the sequential (add-and-shift) approach along with its controller. Compare your design with the array-style multiplier and clearly list the advantages and disadvantages of your proposed design.